

PRE-SEED — DEEP TECH

Planck-99

On-device malware detection built for embedded systems. No cloud. No signatures. No GPU. 37KB of deterministic C that classifies in 34 nanoseconds.

34ns

LATENCY

37KB

FOOTPRINT

96.28%

ACCURACY (IOT)

Zero

NETWORK REQ.

Ziad Salah — Founder & Lead Engineer, 17, Giza, Egypt · Bootstrapped · zs.01117875692@gmail.com · github.com/zierax
All benchmarks public: github.com/Division-36/Planck-99_PublicBenchmarks

Built by someone who learned Linux before he learned algebra.

A decade of deliberate, self-directed work in systems, security, and architecture.

Ziad Salah

Founder · Division-36 · Age 17 · Cairo, Egypt

Self-taught from age 7 — no courses, no bootcamps. Learned Linux from documentation, built a kernel and networking mental model by age 9. Transitioned into offensive security at 12 through independent exploitation work and bug bounty programs.

Ranked **Top 90 globally on HackerOne VDP** (Feb–Jun 2026), #9 Egypt. 12+ critical vulnerabilities in 48 hours.

IEEE AIITA 2026 accepted. **Nature Portfolio** under review by 9 independent experts.

- Age 7 Linux from docs
- Age 12 Bug bounty + exploit research
- Age 15 HackerOne + mentees
- Age 17 Planck-99 + IEEE + Nature



HackerOne VDP — Top 90 Globally

#9 Egypt · 12+ critical vulns in 48h · Feb–Jun 2026 · Publicly verifiable



IEEE AIITA 2026 — Accepted

PTRR Framework · Targeting CSCW on academic reviewer recommendation



Nature Portfolio — Under Review

DVF Framework · 9 independent experts · Scientific Reports



15+ Engines Built Independently

Planck-99, Axiom-WAF, SYRTH, AdmitGPT, and 11 others — no team, no institution

21 billion devices. A hard deadline. Zero solutions.

Traditional AV needs 50–500MB RAM. IoT devices have 256KB–4MB. EU CRA mandates on-device security by August 2027. No production product fits.

21B

IoT & embedded devices globally

Cameras, routers, industrial sensors, medical devices — all connected, all vulnerable, all running on hardware that cannot host traditional security stacks.

IoT Analytics, Oct 2025

200K+

Devices in one nation-state botnet

Flax Typhoon (Raptor Train) compromised cameras, storage, and routers at scale. FBI disrupted it Sept 2024. The playbook is proven and documented.

FBI Director Wray, Sept 2024

0

Commercial on-device classifiers for MCU-class

No production-ready behavioral malware detection engine deploys at 37KB with no network, no GPU, no FPU. The category does not exist yet.

Verified market gap analysis

Structural gap: 256KB–4MB RAM available. AV needs 50–500MB. Cloud needs connectivity. Signatures miss zero-days. **Every existing approach fails the constraint — and regulation now mandates the constraint be met.**

Three forces converging. One has a legal deadline.

Regulation, nation-state activity, and vulnerability rates hitting inflection points simultaneously.

01

EU Cyber Resilience Act — mandatory on-device security by August 2027

Regulation (EU) 2024/2847 entered into force January 2025. Full obligations August 2027. Every device manufacturer selling into Europe must have on-device security. This is a compliance mandate with a countdown, not an opportunity.

EUR-Lex 2024/2847 · Entry into force: Jan 2025 · Full obligations: Aug 2027

02

Botnets are now nation-state weapons — at IoT scale

Flax Typhoon (PRC-state) built Raptor Train from 200,000+ compromised IoT devices. Volt Typhoon sat inside US critical infrastructure for five years undetected. This is geopolitics, not cybercrime.

FBI Director Wray, Aspen Cyber Summit, Sept 2024 · CISA/NSA/FBI Advisory AA24-038A

03

ICS vulnerabilities accelerating — not slowing

2,155 CVEs across 508 ICS advisories in 2025. First year CISA published 500+ ICS advisories. Average CVSS crossed 8.0 for the first time. 75% rated high or critical.

Forescout ICS Cybersecurity 2026, Feb 2026 · SOCRadar ICS Recap, Dec 2025

Every existing solution breaks on embedded hardware.

Benchmarked across every category. Here is what happens when you try to run them on a microcontroller or embedded Linux SoC.

METRIC	TRADITIONAL AV	CLOUD DETECTION	SIGNATURE SCANNER	PLANCK-99 ★
Peak RAM	50–500 MB	Server-side	~30 MB	0.03 MB
Inference time	10–100 ms	200–2,000 ms	1–50 ms	34 ns
Network required	No	Mandatory	No	Never
Air-gap safe	✗	✗	~	✓
MCU-class viable	✗	✗	✗	✓
Deterministic	No	No	~	✓
Accuracy (IoT unseen)	95–98%	N/A	Signatures only	96.28%

★ All benchmarks: github.com/Division-36/Planck-99_PublicBenchmarks — public, reproducible, verifiable · † AV-TEST Institute 2024 · ‡ Competitor specs: publicly documented

Planck-99

32-dimensional normalized frequency vectors. Int8-quantized closed-form inference. No neural networks. No cloud. No GPU. No FPU. No runtime dependencies.

DETERMINISTIC

34 ns

Median inference on IoT syscall traces.
 $\sigma \leq 14$ ns across 9 independent runs.
Same input, same output, every time.

ULTRALIGHT

37 KB

27 KB int8-quantized binary + 10 KB
ring buffer. Fits in L1 cache of any
embedded Linux SoC. No AI accelerator
required.

ACCURATE

96.28%

On unseen IoT malware 2016–2026
(ADFA-LD). Real generalization across
10 years of threats. 97.71% precision.

THROUGHPUT

29.4M/s

Inferences per second on a single CPU
core. No batching, no GPU. Pure
deterministic dot product on a 32-
feature vector.

Verified on: Intel Xeon 3.8GHz, single core, no GPU, no network · github.com/Division-36/Planck-99_PublicBenchmarks

Closed-form syscall analysis. Zero black boxes.

Training computes a mathematical decision surface. Inference is a quantized dot product. Every classification produces a JSON proof file.

Math, not magic

Most ML in security is a black box. Feed it data, get a probability, hope it's right. **Planck-99 is different.**

We model syscall sequences as **32-dimensional normalized frequency fingerprints**.

Training computes a closed-form decision surface. At runtime, inference is a deterministic int8 dot product — identical input always yields identical output.

The model generalizes 51× beyond its training ceiling (tested to 117,088 syscalls) because normalized frequency ratios are length-invariant by construction. No stochastic behavior. No runtime training.

- 1 Kernel hook captures syscall trace into ring buffer
- ↓
- 2 Extract 32 normalized frequency ratios (closed-form)
- ↓
- 3 Int8 deterministic classification (dot product)
- ↓
- 4 JSON proof file generated — CRA Article 13 audit trail

It works on data it has never seen.

All benchmarks public. No cherry-picking. IoT dataset: malware 2016–2026 never seen during training.

BASELINE

Training Set (Internal Validation)

ACCURACY	PRECISION
98.88%	98.58%
RECALL	FPR
99.21%	1.45%

UNSEEN

LinuxStatic ELF Dataset

ACCURACY	PRECISION
92.01%	100%
RECALL	FPR
92.01%	0.00%

UNSEEN — PRIMARY TARGET

IoT Syscalls (ADFA-LD)

ACCURACY	PRECISION
96.28%	97.71%
RECALL	FPR
97.87%	12.18%*

Absolute kernel latency — deterministic, trace-length invariant

Training validation traces (median 863 syscalls)	62 ns $\sigma \leq 14$ ns
LinuxStatic ELF (median 1,161 syscalls)	35 ns $\sigma \leq 14$ ns
IoT Syscalls (median 15,075; max 117,088)	34 ns 51× training ceiling extrapolation

* IoT FPR 12.18%: data-distribution artifact (1:5.3 benign:malware ratio, short benign traces). Fully mitigated by 500-syscall gate in production. Internal FPR on balanced dataset: 1.45%.

\$28B → \$80B by 2031

18.7% CAGR. Real driver: 2027 compliance cliff, not organic growth.

\$80B

Global IoT Security Market, 2031

● EU CRA: Forced Retrofit of 21B Devices

Every connected device sold in Europe must meet mandatory cybersecurity requirements by August 2027. Manufacturers who cannot demonstrate on-device security will be legally barred from the market.

EUR-Lex 2024/2847 · Technavio, March 2026

● Industrial OT: Air-Gapped by Definition

Industrial OT requires on-device, air-gapped detection. No cloud solution qualifies. Industrial IoT security alone: \$26.24B in 2024. North America: 34.2% global share.

Technavio, March 2026

● APAC: Highest-Growth Region

APAC IoT security CAGR: 22.2% — highest globally. Japan JC-STAR (2024) signals regulatory alignment. Rapid industrial digitalization across Southeast Asia and India.

Technavio, March 2026 · Japan MIC JC-STAR, 2024

● Regulatory Stack: CRA, PSTI, NIST, FDA

EU CRA (2024/2847). UK PSTI Act (April 2024). FDA Cybersecurity Guidance (2023). NIST SP 800-213A. Every regulation mandates on-device security. Compliance is not optional.

EUR-Lex · legislation.gov.uk · FDA.gov · NIST

Three phases. Illustrative unit economics.

No revenue projections — fantasies until a contract is signed. Directional priorities aligned with EU CRA 2027 window.

DESIGN PARTNER PHASE	OEM PLATFORM DEAL	CHIPSET EMBED (NXP/STM)
50K–500K	1M–10M	100M+
devices · \$0.10–\$0.50 per device license	devices · \$0.05–\$0.20 per device (volume)	devices · \$0.01–\$0.05 royalty per unit
\$5K–\$250K	\$50K–\$2M	\$1M–\$5M
ILLUSTRATIVE ARR	ILLUSTRATIVE ARR	ILLUSTRATIVE ARR AT SCALE

~100% gross margin. Pure software SDK — no hardware, no manufacturing, no COGS. Scales linearly with device count. Pricing validated with first design partner.

Phase 1 target design partners (illustrative — no signed agreements):

- Advantech
- Kontron
- Moxa
- Red Lion Controls
- Toradex
- NXP (Phase 2)
- STMicroelectronics (Phase 2)
- Espressif (Phase 2)

We're not improving a category — we're creating one.

No competitor checks all four boxes. Most fail on two or more. Category gap legally mandated by 2027.

CATEGORY	ON-DEVICE	AIR-GAP SAFE	REAL-TIME	MCU-CLASS	NOTES
Traditional AV (Kaspersky, CrowdStrike)	✗	✗	✓	✗	50–500 MB RAM requirement
Cloud Detection (Darktrace, Vectra)	✗	✗	~	✗	Mandatory connectivity
Signature Scanners (YARA, ClamAV)	~	✓	✗	✗	No behavioral analysis
OT Platforms (Nozomi, Claroty)	✗	~	~	✗	Network layer only
Planck-99	✓	✓	✓	✓	37KB · 34ns · 96.28%

★ [github.com/Division-36/Planck-99_PublicBenchmarks](#) · † AV-TEST Independent Test 2024 · ‡ Competitor features: publicly documented specs

One design partner. That's the only milestone that matters.

Everything else is noise until a real customer validates the integration. Bootstrapped. Open to pre-seed partners and strategic investors.

1

Find the right vendor

An IoT gateway or OT security vendor shipping embedded Linux devices. Advantech, Moxa, Kontron class. The product already runs on their hardware. Integration, not invention.

2

Run a live benchmark

Planck-99 deploys as a 27KB binary with a kernel hook. Integration takes hours, not months. Vendor gets a JSON proof file per classification — auditable, CRA Article 13 compliant.

3

Sign a paid pilot

A signed agreement is the proof of market demand no deck can substitute. De-risks every subsequent conversation with investors, accelerators, and OEM partners.

On single-founder risk: Acknowledged. The core engine is 37KB of deterministic C — not team-dependent. IP and architecture fully documented. Execution risk is commercial, not technical.

zs.01117875692@gmail.com github.com/zierax linkedin.com/in/z14d Raising: \$1M pre-seed

All benchmarks public: github.com/Division-36/Planck-99_PublicBenchmarks